

Augmented reality, deep learning and vision-language query system for construction worker safety

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ARTICLE INFO

Keywords:

Construction safety
Deep learning
Vision-language models
Augmented reality

ABSTRACT

Low situational awareness contributes to safety incidents in construction. Existing Deep Learning (DL)-based applications lack the capability to provide context-specific and interactive feedback that is essential for workers to fully understand their surrounding environments. This paper proposes the Visual Construction Safety Query (VCSQ) system. The system encompasses real-time Image Captioning (IC), safety-centric Visual Question Answering (VQA), and keyword-based Image-Text Retrieval (ITR), integrated with head-mounted Augmented Reality (AR) devices. System validation includes benchmarks and real-world images. The ITR module posted high recall rates of 0.801 and 0.835 for Recall@5 and @10. The VQA module achieved an 89.7% accuracy rate, and the IC module had a SPICE score of 0.449. Feasibility tests and surveys confirmed the system's practical advantages in different construction scenarios. This study establishes an integration roadmap adaptable to future advancements in interactive DL and immersive AR.

1. Introduction

Ensuring worker safety is an essential aspect of construction management. Despite the provision of safety measures, statistics reveal that unsafe practices persist [27]. For instance, the Australian construction sector reported 183 fatal injuries from 2017 to 2021, which highlights the severity of the issue [2]. In developing countries, the situation is even worse. Between 2009 and 2019, China recorded an average of 2851 fatalities per annum due to non-compliance with safety regulations [1]. To mitigate the safety risks, construction workforce needs thorough upskilling and re-skilling. Nevertheless, traditional construction safety methods might rely too much on passive approaches, such as signage, periodic safety inspections, and verbal briefings. While these methods can be useful for conveying safety information, they may not be effective in promoting long-term retention of safety knowledge and reinforcing situational awareness [2]. In addition, these approaches lack the necessary interaction between safety instructors and workers, which can help to clarify any confusion and answer specific safety-related questions.

Active learning approaches involving critical thinking, exploration, and conversation about safety regulations have been introduced to improve construction safety [35,42]. Active learning can also be facilitated by innovative visualisation methods such as Virtual Reality (VR) [20,22,46], Augmented Reality (AR) [5,8], expert enquiry systems that enable workers to seek clarification on safety concerns [39,50]. Pre-defined AR/VR scenarios can aid workers in comprehending safety regulations through step-by-step guidance, alerts, and reminders. However, introducing real-life experiences into active learning through these digital approaches has never been an easy undertaking. Expert query systems allow workers to communicate with Artificial Intelligence (AI) via natural language and to quickly access critical safety information and guidelines [39]. Although text-based expert query systems can offer valuable information to onsite workers, the generated textual content can be compromised due to the subjective and incomprehensive inputs provided by workers. For example, a worker might provide a description of their proximity to a machine, but fail to mention other important factors, such as the machine statuses or the worker's PPE. This can result in the system failing to identify potential safety hazards or

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<https://doi.org/10.1016/j.autcon.2023.105158>

Received 9 April 2023; Received in revised form 19 October 2023; Accepted 26 October 2023

Available online 31 October 2023

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providing inaccurate safety information.

Several researchers have concentrated on Deep Learning (DL)-based vision-language models that incorporate image inputs to complement and enhance the understanding of construction site [16,49]. These models typically involve an encoder to process visual information, a decoder to generate textual descriptions, and appropriate loss functions to optimise the learning process. Bang and Kim [4] introduced a dense captioning method to generate explanatory texts, encompassing information on the position, status, movement, colour, and quantity of construction resources in drone-captured images. Liu et al. [28] proposed a DL-based algorithm for efficiently manifesting the activity scenes of construction site images with complete sentences. Kim and Chi [23] utilised a novel graph neural network for detecting visual relationships among construction resources. Their study enhanced construction scene understanding by considering the propagation effects of various object interactions. Current vision-language models in construction can effectively extract high-level visual cues and provide workers with informative captions regarding construction activity states. However, they may not always be able to address the specific safety concerns or respond to in-depth inquiries based on pictures. This limits the ability of workers to receive tailored and accurate safety guidance in real-time.

To bridge the gaps mentioned above, this study introduces an interactive AR/DL-based query system, named Visual Construction Safety Query (VCSQ), for providing on-demand safety information by processing on-site images. Our contributions include:

1. Proposing a unified bidirectional image-text processing approach, which incorporates natural language instructions and leverages a pre-trained multimodal mixture of encoder-decoder (MED) model. The model includes three key functionalities: Image Captioning (IC) for comprehensive scene descriptions; Visual Question Answering (VQA) for answering queries related to image content; and Image-Text Retrieval (ITR) for accurate alignment of image content with corresponding text.
2. Introducing an innovative prompt-tuning method that exploits the zero-shot capability of the pre-trained model. This approach facilitates rapid adaptation to safety-related context, thereby increasing the versatility and generalisation of the VCSQ system.
3. Incorporating AR experiences into an interactive interface for the VCSQ system. This integration reduces the reliance on pre-defined scenarios and allows for more natural interactions between workers and the system.

The remainder of this paper is structured as follows: Section 2 explores existing applications of DL in construction and methods of AR visualisation. Section 3 elaborates on the methodologies employed. Section 4 presents the results of the experiments conducted. Section 5 evaluates the practicality of the system. Section 6 discusses limitations and outlines directions for future research. Finally, Section 7 provides the conclusions.

2. Literature review

Construction scene understanding using computer vision and natural language processing is worth investigating in the realm of construction safety management. This section examines DL-based methodologies in construction scene understanding, their pros and cons, DL-supported AR technology knowledge, research gaps, so on so forth.

2.1. DL-based methods for construction safety

To ensure construction site safety, computer vision technology has been widely used for construction safety applications, such as Personal Protective Equipment (PPE) detection, proximity detection and mobile plant tracking [7,18,29,37]. However, these applications may not fully capture the complex interdependencies among construction activities,

worker statuses, and construction elements, which is important to have a comprehensive understanding of potential hazards. Benefiting from recent advances in DL, vision-language models open up a new avenue for construction scene understanding. Vision-language models, which integrate visual and textual information, are designed to process and interpret multi-modal data. As a result, they can effectively analyse images and generate natural language outputs, which allows for a more accurate representation of the complex relationships between dynamic objects in construction environments. A typical vision-language application is IC, which is used to provide the comprehensive explanatory text of construction images. Liu et al. [28] adopted IC to depict construction worker activity scenes and proposed a linguistic schema for annotating construction images. Bang and Kim [4] applied IC to drone-acquired images for vision-based monitoring of construction sites, achieving a Mean Average Recall (mAP) of 45.52%. Xiao et al. [45] conducted a feasibility study of IC in construction, in which they compared and analysed six IC-based models. However, the use of generic image captions may not significantly contribute to construction safety. To better address the specific needs of the construction safety domain, Li et al. [27] developed a visual relationship detection model for onsite hazard identification based on ontology. Chen et al. [10] presented a scene graph structure that identify improper PPE wearing based on spatial relationship of PPE and workers. Pan et al. [32] utilised a knowledge graph-based DL model to interpret construction images based on specific safety rules. Fang et al. [16] integrated a knowledge graph for hazard identification. Their development can identify fall-from-height hazards. Although these vision and language-based applications can generate descriptive captions to identify unsafe behaviours, their interactivity with workers has been limited. Without the ability to ask specific questions or receive personalised guidance, workers may struggle to effectively utilise safety knowledge, particularly in unique situations where specific guidance is required. Thus, it is worth exploring more advanced vision-language models that incorporate interactive features, enabling workers to actively engage with the system, ask context-specific questions, and receive accurate, real-time safety information.

2.2. AR-based visualisation techniques

The construction industry has been increasingly utilising advanced visualisation technologies to automate information displays [43]. AR technology enables users to view and interact with holograms, which are objects made of light and sound that resemble real objects [44], while still maintaining their connection to the real world [6]. An interview made by Harikrishnan et al. [17] demonstrated that AR has the potential to assist in inspection and query. Chen et al. [11] utilised holograms to visualise fire safety-related files and drawings for inspectors to ensure fire safety equipment is in good working condition. Chi et al. [12] integrated AR and laser scanning to provide an intuitive and accurate method to check the number, spacing, and location of rebars, simplifying rebar installation inspection. Numerous studies have also examined the application of AR devices for upskilling and investigating the learning effects. Pilati et al. [33] found that the use of AR effectively reduced training time and enhanced worker performance in assembly tasks. Wang et al. [41] and Kim et al. [24] conducted experiments to evaluate AR-based training materials, demonstrating their superiority over text-graph-based materials for spatial skills learning. Eiris et al. [15] and Ogunseju et al. [30] illustrated the benefits of AR in hazard identification training and personalised posture training, leading to improved workforce safety and health. These investigations have yielded insightful findings on aspects such as learning motivation, knowledge transfer, and cognitive load. DL and AR have been integrated to assist in hazard identification and precautions. Kim et al. [25] developed a DL-based construction hazard avoidance system on AR goggles to provide users with the directions of hazard sources through holographic arrows. Wu et al. [44] integrated DL and AR to develop a visual warning

system that can show hazard sources, highlight dangerous areas, and provide proximity alerts. These studies focus on visualising the hazard information through symbols (arrows and signs) and 3D models (virtual vehicles and workers), which enable users to intuitively identify surrounding hazards. DL algorithms in these studies were used to detect and calculate the proximity of the hazard sources and workers. The corresponding positioning signals were then transmitted to a digital replica, which can be displayed through AR. While these visualisations are valuable for hazard awareness, they often lack the provision of contextual information such as safety guidelines or situational advice. This gap in contextual understanding limits the efficacy of AR-based solutions in fully informing workers for safer decision-making on construction sites. The integration of advanced image-text processing capabilities into AR systems could significantly improve this aspect, enabling workers to understand not just what the hazards are, but also how best to approach or avoid them. To date, such an integration is scant but worth of investigating.

2.3. Research gaps and objectives

Existing vision-language models employed within the construction safety domain may not accurately deliver the precise details or context-specific insights that workers seek. This deficiency in interactive capabilities could constrain workers' comprehension of safety regulations in particular situations. Current construction safety applications that integrate DL with AR do not consider the communication requirement of users which makes it challenging to understand the surrounding environment. The interactivity and effectiveness of these applications in promoting safe actions require further enhancement. To address the identified gaps, this research aims to develop a comprehensive, adaptable, and interactive vision-language model integrated with AR technology for construction safety applications. The two key objectives of this study are as follows:

1. Developing a multi-functional safety query system that leverages pre-trained vision-language models and prompt design methods, providing a tailored experience that addresses individual worker needs, enables a thorough understanding of safety regulations, and facilitates real-time interaction with on-site workers.
2. Introducing an integration of a vision-language model within AR to offer context-aware safety information and guidance for construction workers on-demand. This integration allows workers to receive tailored safety knowledge based on their specific location and tasks, enhancing their understanding of the potential safety issues and promoting safer practices in real-world situations.

3. Methodology

The VCSQ system consists of three main modules: IC, VQA, and ITR, all built upon a pre-trained vision-language model. A pre-trained model refers to a model that has been previously trained on a large dataset containing various images and associated textual information. This enables the model to understand and extract patterns from both visual and textual data, making it suitable for vision-language tasks. To ensure the provision of safety-related answers, prompt-tuning strategies are incorporated to adapt the pre-trained model to the specific context of construction safety regulations. Upon receiving a construction image, the IC module generates descriptive captions, enabling workers to comprehend the site environment. The VQA module features a chatbot, allowing workers to use voice input via AR glasses to inquire about safety compliance, thereby enhancing their understanding of safety regulations and requirements. The ITR module is designed to help workers or managers quickly locate and assess the safety status of specific objects on the construction site using keyword searches. This comprehensive system aims to elevate the safety awareness of construction workers by providing nuanced insights into the specific

hazards present on construction sites. The following subsections describe IC, VQA, and ITR, and AR integration in detail, as well as the implementation of prompt-tuning strategies for tailoring the model to the construction safety domain.

3.1. Structure of the pre-trained model

Our research aims to develop new applications for construction safety by leveraging pre-trained object detectors and Large Language Models (LLMs).

We are particularly interested in the potential of LLMs to facilitate visual knowledge reasoning and to enable interactive visual interfaces [36]. Given that LLMs predominantly focus on language and often do not directly process image data, integrating visual data into a feature space comprehensible by LLMs is a challenge. While traditional encoder-decoder models excel at single-task objectives, they may not be as effective for multi-task applications that require the simultaneous processing of visual and textual data [19]. Unlike traditional encoder-decoder models, CLIP [34] is trained in a way that allows it to handle multiple types of vision-language tasks without task-specific training data, making it a more flexible and generalized model for multi-modal tasks. Bootstrapping Language-Image Pre-training (BLIP) framework [26] advances over CLIP by being designed for unified vision-language understanding and generation, achieving state-of-the-art performance across various vision-language tasks. BLIP is built upon a MED structure, designed to handle multiple vision-language tasks concurrently. MED is designed to simultaneously handle multiple tasks by processing both visual and textual data within the same model architecture. This capability makes MED and by extension, BLIP, ideally suited for our complex research objectives, which encompass tasks like IC, VQA, and ITR.

The BLIP framework is tailored for streamlined pre-training across various tasks and enables flexible transfer learning between different types of data. A Vision Transformer (ViT) [14] (Fig. 1) was employed as the image encoder. ViT is a transformer-based architecture that has shown great potential in computer vision tasks. The architecture is designed to utilise the powerful multi-head self-attention mechanism for image processing. By employing the Transformer architecture, ViT can capture both local and global information through self-attention, which makes it a promising alternative for computer vision tasks that require processing large amounts of visual data. The input image is first resized and divided into non-overlapping patches (e.g., 16×16 pixels). Each patch is then flattened into a one-dimensional vector and linearly embedded using a fully connected layer to create a fixed-size input vector for the Transformer. Positional information is added to each patch using position embeddings since the transformer architecture is not inherently sensitive to spatial positions. These embeddings are learnable parameters initialised randomly and are added to the corresponding patch embeddings. The output of the previous step is used as input to a series of Transformer encoder blocks. Each block comprises two main sub-layers, which are multi-head self-attention and feed-forward layers. Multi-head self-attention computes the attention scores between every pair of patches (including the patch with itself) in the input sequence. Feed-forward layers are the Multi-Layer Perceptron (MLP) layers which consist of two fully connected layers with a non-linear activation function in between. The feed-forward layers are responsible for capturing non-linear relationships within the data. Each sub-layer within the Transformer encoder block is surrounded by layer normalisation and residual connections to stabilise the learning process and facilitate deeper architectures. After passing through the series of Transformer encoder blocks, the output corresponding to the first token (usually a special [CLS] token) is used to make the final prediction. A fully connected layer (the MLP head) followed by a SoftMax activation function generates class probabilities for the image encoding process.

The MED architecture can perform a variety of image-text generation tasks, including IC, VQA, and ITR, utilising its three encoder-decoder structures. As shown in Fig. 2, unimodal encoder is similar to

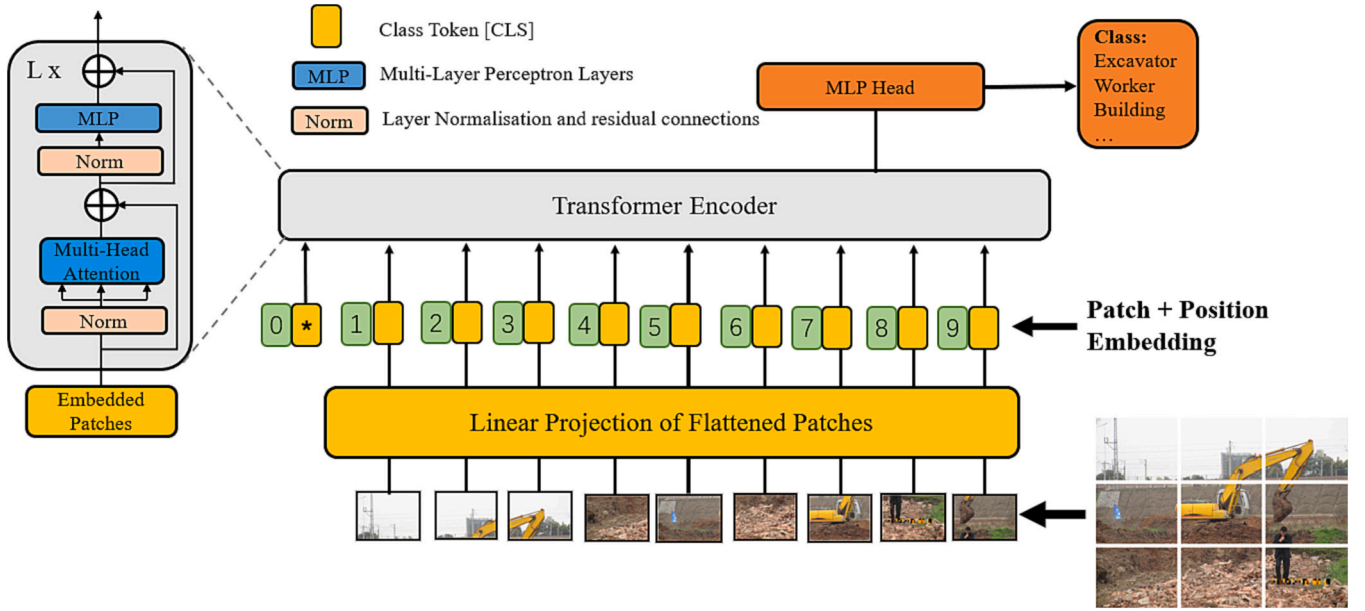


Fig. 1. Architecture of image encoder.

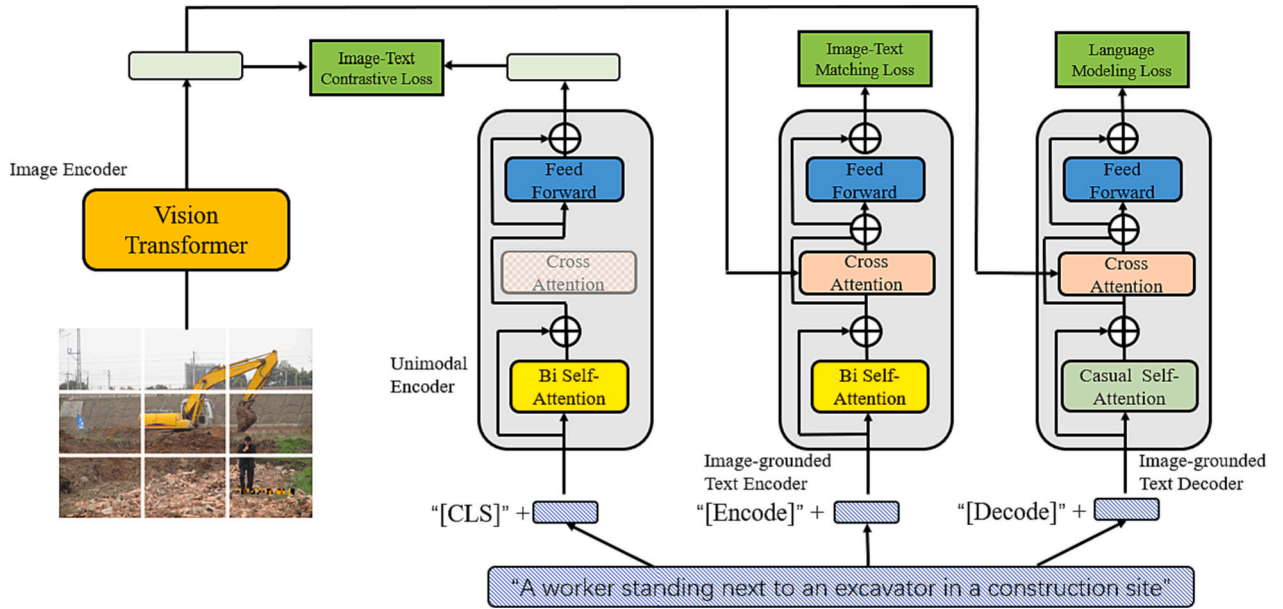


Fig. 2. Overall structure of MED.

Bidirectional Encoder Representations from Transformers (BERT) [13], with a [CLS] token added to the beginning of the text input to summarise the sentence. The unimodal encoder was trained using an image-text contrastive loss function. This loss function helps align the feature spaces of the visual and text transformers. The resulting encoder-decoder structure can perform ITR tasks that involve comparing or matching image and text features separately.

The image-grounded text encoder incorporates visual data through introducing a cross-attention layer between the feed-forward network and the self-attention layer within each transformer block of the text encoder. The text is appended with a task-specific [Encode] token. This token is used to indicate to the model that the following input sequence will be combined with image features, and the output embedding of this token serves as the combined representation of the image-text pairs. The image-grounded text encoder was trained using an image-text matching

loss function. The purpose of this function is to learn multimodal representations that capture the precise alignment between vision and language. This encoder incorporates visual information into the text encoder, making it ideal for VQA tasks that require understanding and reasoning over both image and text inputs.

The image-grounded text decoder employs causal self-attention layers instead of bidirectional self-attention layers. A [Decode] token is used to signal the beginning of a sequence and tell the model that it should start generating the text output from the decoder. The image-grounded text decoder generates textual descriptions from input images. It utilises causal self-attention to predict tokens in an autoregressive manner, ensuring coherent captions. The language modelling loss function is optimised by minimising cross-entropy loss, which maximises the likelihood of the text. By conditioning text generation on image features, this decoder is designed for IC tasks.

3.2. Pre-training and prompt-tuning strategies

A significantly larger number of image-text pairs are utilised in recent studies, and these pairs are automatically obtained from the web to pre-train extensive models [34]. Nevertheless, the image-texts frequently fail to provide an exact depiction of the visual contents within the images, resulting in a noisy signal that is less than ideal for mastering vision-language alignment. BLIP has been pre-trained on the image-text pairs collected from the web by utilising a data augmentation method called Captioning and Filtering [26], which can enhance the quality of the text corpus. To generate captions for web images and filter out noisy image-text pairs, this method initialises both the captioner and the filter from the same pre-trained vision-language model and fine-tunes them on the public dataset. The filtered image-text pairs, along with manually annotated pairs, can be combined to create a new dataset that can be used to pre-train the model.

As construction safety vision-language tasks often involve specialised terminologies and unique visual patterns, which may not be well-represented in the pre-training data. Prompt design can be used to adapt the models to the specific domain with a small amount of task-specific data. Recent approaches to VQA tasks involve treating them as multi-label classification tasks [38,40]. However, the difficulty lies in designing prompts for VQA tasks since the text inputs used for VQA (questions) and pre-training tasks (captions) differ in form. In order to address this issue, we adopted a generation approach [21] for VQA tasks to avoid the difficulty. In this configuration, prompts act as a form of guidance that direct the models towards producing answers in the desired format. In the absence of prompts, the models may produce VQA sentences that do not adhere to the correct format. To this end, we explored various input and output prompts, which are detailed in Tables 1. The token *[Q]* and *[A]* denote the question-and-answer text, respectively. The token *<text_1>* serves as a sentinel and is included in the input sequence to signify the beginning or end of a sequence. It is often used to help the model learn where a sentence or sequence begins and ends and is not present in the actual text. Regarding input prompts, we explore three different templates, namely, “question: *[Q]* answer:”, “*[Q]* <text_1>”, and “question: *[Q]* answer: <text_1>”. Two distinct formats were employed for output prompts, specifically “*[A]*” and “<text_1> *[A]*”.

Construction safety VQA tasks may include two types of questions: open-ended questions and closed-ended questions. Closed-ended questions have a finite number of possible answer choices, while open-ended questions do not. To this end, we convert questions into statements using a technique called Question-to-Statement (Q2S) prompt [51]. When questions are converted into statements, they become more like captions or descriptions of the image, which is the type of input that the model was pre-trained. Specifically, we apply Image-Text Matching (ITM) for closed-ended questions and Masked Language Model (MLM) for open-ended questions. Specifically, closed-ended questions can be formulated as a classification task. ITM involves learning to match image features with corresponding text features, which enables the system to determine the correct answer choice from a pre-defined set of options. When dealing with a closed-ended question such as “Are any workers

wearing harnesses?”, the prompt statement is created as “Some workers are wearing harnesses.”. The ITM head takes in the text embedding and visual features and predicts if they match. A high match score implies a “Yes” answer, while a low score indicates “No”. For open-ended questions, MLM is used because the answer is not limited to a pre-defined set of choices. In MLM, a certain percentage of the input text is randomly masked, and the model is trained to predict the correct missing words. This allows the system to generate a diverse range of answers to open-ended questions. For an open-ended question like “How many workers in the image?”, the prompt is generated as “There are [MASK] workers.”. The [MASK] token is a placeholder for a word or phrase that the model is expected to predict or fill in based on the context provided in the prompt. In this case, we identify the most appropriate answer for a given question within a set of candidate answers. To accomplish this, we search the possible answer set to identify the answer with the greatest probability. By using the Q2S prompt technique, we can effectively adapt pre-trained models to answer construction safety regulation questions based on visual data, enhancing safety management and compliance in the construction industry.

To ensure accurate and safe guidance, our VCSQ system, implemented within AR glasses, employs an approach to tackle the issue of erroneous or made-up answers. Given that the audio system in the AR glasses can misinterpret verbal input and introduce noise, we incorporate an uncertainty prompt “If you are not sure about the answer, say uncertain.” Acknowledging uncertainty is critical in a construction safety context, as the repercussions of incorrect advice can be severe. This not only safeguards against potential hazards but also fosters trust with the workers by not misleading them. To further stress-test the model’s response mechanism, we include irrelevant prompts, random sentences, and noisy tokens, as detailed in Table 2. The model is trained to reply ‘Uncertain’ when faced with these elements, thus providing a robust response rather than incorrect or irrelevant answers.

3.3. Development of the AR features

The architecture of the developed AR features is demonstrated in Fig. 3, where Microsoft HoloLens 2 are employed as display devices. The spatial mapping function in HoloLens facilitated the creation of a convincing AR experience, enabling holograms to appear real. Unity 3D was utilised to build human-machine interfaces and network connection functions. C# scripts were developed to achieve hand tracking, voice input, and gaze input, providing natural interaction methods. To connect the AR components to the DL modules, a server-client architecture was applied. A computer running a C# program was the server. The AR components, the DL modules, and a Closed-Circuit Television (CCTV) were the clients. The server and clients were connected using Transmission Control Protocol (TCP) sockets. Real-time information, such as data, texts, and images, was encoded into byte arrays and delivered between the server and clients through the sockets.

An AR User Interface (UI) was created to enable user interaction with the VCSQ system. As depicted in Fig. 4, the UI is divided into three

Table 1
Hand-crafted prompts.

Task	Input prompt	Example
VQA	<i>[Q]</i> <text_1> input:	What safety equipment is this worker wearing? <text_1> output: <text_1> hard hat, safety vest
	question: <i>[Q]</i> answer:	input: question: What safety equipment is this worker wearing? answer: output: <text_1> hard hat, safety vest
	question: <i>[Q]</i> answer: <text_1>	input: question: What safety equipment is this worker wearing? answer: <text_1> output: <text_1> hard hat, safety vest

Table 2
Prompt categories.

Input prompt template	Category
<i>[Q]</i> What is the date today according to the calendar?	irrelevant prompts
Fill in the missing word in this sentence: <i>[Q]</i>	
What is your favourite movie? <i>[Q]</i>	
<i>[Q]</i> noisydata	noisy tokens
rubbishdata appeared on the screen. <i>[Q]</i>	
<i>[Q]</i> Can you sing the alphabet backwards?	
<i>[Q]</i> A worker is wearing a hard hat and safety vest.	random sentences
<i>[Q]</i> A crane is lifting heavy materials at a construction site.	
A worker is using a jackhammer to break concrete. <i>[Q]</i>	
A group of construction workers discussing project plans. <i>[Q]</i>	random sentences
<i>[Q]</i> A construction worker is operating an excavator.	

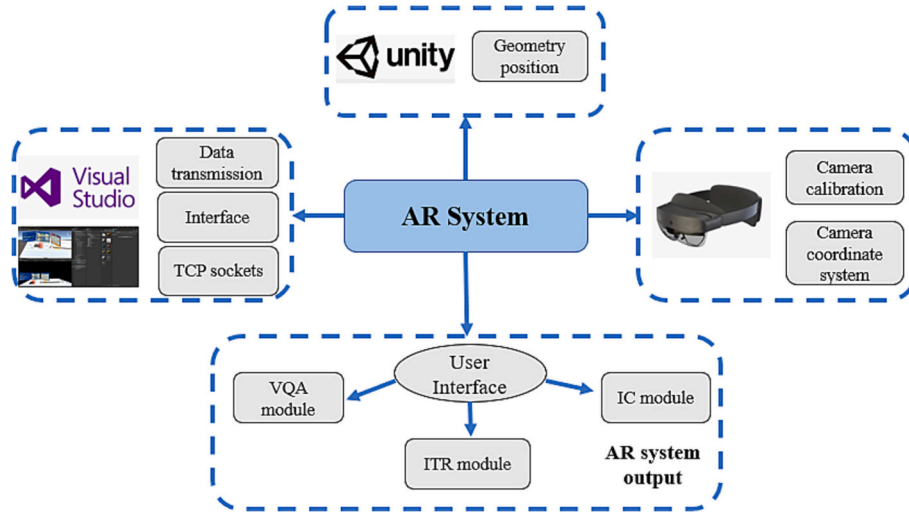


Fig. 3. Structure of the AR-based module.

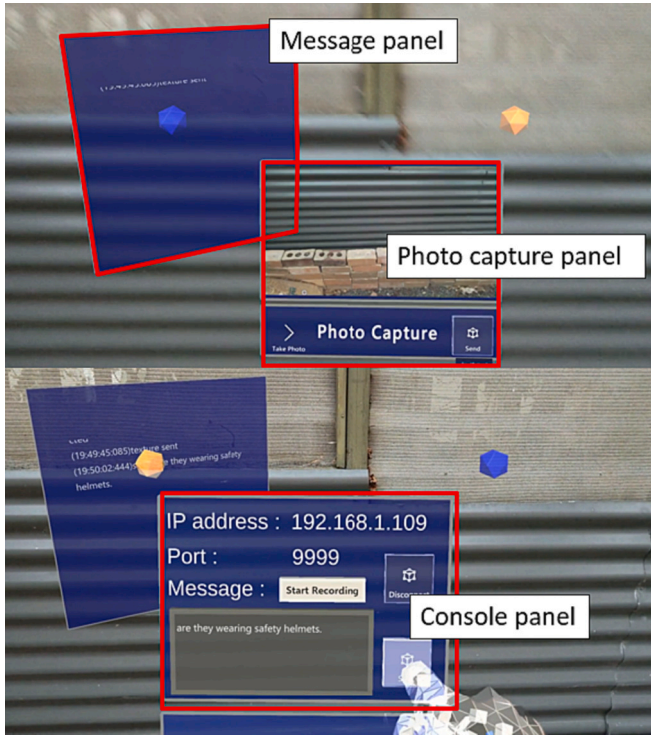


Fig. 4. Interface of the AR application.

sections: a photo capture panel, a message panel, and a console panel. Users have the option to show or hide these panels either by focusing their gaze on triggers marked by blue and orange gems or by vocalising specific keywords ('photo capture', 'message', and 'console'). The photo capture panel allows users to take real-time photos through CCTV or activate the photo capture function, which are then sent to the VCSQ system for processing. Featuring a speech-to-text functionality, the message panel lets users verbally pose questions to the query system and receive answers. These answers and activity logs are displayed on the console panel. To reduce cognitive load and address AR occlusion issues, the AR interface includes hide/unhide buttons. These buttons allow users to minimise text overlays and occlusion. The AR interface thus offers an intuitive platform for users to seamlessly interact with the query system, enabling more informed decision-making in various real-

world construction scenarios.

4. Implementation, experiments, and results

As the development in this study is a proof-of-concept prototype, the range of the query system is limited to hazardous circumstances associated with 7 common construction safety considerations (Table 3). Each example question is directly linked to these safety considerations. These questions and their respective answers will later form the basis for dataset development. To broaden the scope of safety considerations beyond the initial seven categories, one effective strategy could be the use of LLMs like GPT-4 [31]. Utilising such models can assist in generating a range of questions that address diverse safety issues, thereby enriching the dataset. This approach could significantly expedite the task of compiling a more comprehensive and relevant set of safety questions. The experiment is conducted on a server with a 3.40 GHz Intel Core i7-13700KF CPU and GeForce RTX 4080 GPU with 16 GB VRAM. The proposed system has been implemented with Python language. The PyTorch framework was used to perform the VCSQ system.

4.1. Data collection and pre-processing

Due to the absence of a publicly available construction safety VQA dataset, we have undertaken the task of creating our own dataset, focusing on construction safety scenarios. The dataset captures various aspects such as workers approaching hazardous areas, engaging in unsafe behaviours, and different environmental factors such as illumination, scale, and occlusion. In total, we have gathered 578 images and

Table 3
Safety considerations investigated in this study.

ID	Safety Considerations	Related Questions
1	Scaffold Safety Standards	Are there proper guardrails and toe boards on the scaffolding?
2	Excavation Barrier Requirement	Are the excavation sites surrounded by clearly visible barriers or barricades?
3	Onsite Worker Protection Gear	Are workers wearing safety helmets and high-vis harnesses properly?
4	Height Work Harnesses Mandatory	Are workers wearing safety harnesses when working at heights?
5	Clear and Obstruction-Free Exits	Are emergency exits and escape routes clearly marked and unobstructed?
6	Adequate Site Lighting Required	Is the construction site equipped with adequate lighting?
7	No Smoking Permitted Anywhere	Can you spot any workers smoking on the construction site?

corresponding question-answer pairs for fine-tuning and testing purposes, with some images encompassing multiple unsafe behaviours. The images and question-answer pairs were partitioned into training, validation, and testing sets in an 8:1:1 ratio, according to the type of unsafe behaviour and corresponding textual description.

The primary source of construction images was the large-scale construction image dataset, Moving Objects in Construction Sites (MOCS) [47]. During the data selection process, we considered various factors such as illumination, size, and occlusion to mitigate potential overfitting issues. As the MOCS dataset was designed exclusively for object detection and segmentation, additional pre-processing was required. This included generating corresponding question-answer pairs with ample information to accurately assess construction safety scenarios.

An example of the VQA annotation is depicted in Fig. 5. For each annotated image, an .JSON file is generated, encompassing the query text (questions), correct answers, and any additional information relevant to the construction safety scenario. We have developed the dataset in a standard format to cater to different requirements and ensure compatibility with the majority of existing VQA learning models. This newly created construction safety VQA dataset enables researchers and practitioners to train and evaluate models tailored to the specific domain of construction safety.

4.2. Parameter setting

For the fine-tuning process of the VQA module, we first resize all the images into 256×256 . Since construction images are often suffered from occlusions, which could lead to inaccurate detection results. To this end, we perform an effective data augmentation method proposed in [9] to improve the quality of the images. During the training process, the batch size is set to 8 and Training is conducted over 100 epochs with a learning rate of 0.001, halved every 10 epochs.



Fig. 5. VQA data format.

4.3. Performance evaluation of the vision-language tasks

A variety of metrics were used to validate the accuracy of the development in various vision-language tasks. To begin with, PASCAL VOC metrics have been implemented to evaluate the accuracy performance of object detection, referred to as the image encoding process. The Intersection over Union (IoU) can be computed by dividing the area where a predicted box (B) and a ground truth bounding box (B_{gt}) overlap by the area of their combined space:

$$IoU = \frac{Area(B \cap B_{gt})}{Area(B \cup B_{gt})} \quad (1)$$

In order to qualify as a True-Positive (TP) detection result, three conditions must be met: a confidence score above 0.7; accurate classification of the predicted category; and an IoU value greater than or equal to 0.5. If the latter two criteria are not met, the result is classified as a False-Positive (FP). The PASCAL VOC metrics also make additional assumptions for determining TPs and FPs, such as selecting only the bounding box with the highest confidence score if the ground truth is linked to multiple boxes. Similarly, if the confidence score falls below a certain threshold, the result is considered a False-Negative (FN). Based on these criteria, Precision and Recall can be calculated using following specific formulas:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Where, Precision is calculated as the proportion of TPs to the sum of TPs and FPs, and Recall is the proportion of TPs to the total number of target objects. Table 4 demonstrates the detection performance of the

Question.JSON

```
{
  "1238592": {
    "imageId": "2407890",
    "question": "Are these workers wearing Personal protective Equipment properly?",
    "answer": "yes",
    "fullAnswer": "Yes, the workers are wearing Personal protective Equipment.",
    "isBalanced": true,
    "group": "8r-binary-PPE",
    "entailed": ["1352631", "1245832", "842753"],
    "equivalent": ["1245832", "842753"],
    "types": {
      "structural": "verify",
      "semantic": "relation",
      "detailed": "existAttrRel"
    },
    "annotations": {
      "question": {"4": "271881", "7": "279472"},
      "answer": {},
      "fullAnswer": {"4": "271881"}
    },
    "semantic": [
      {
        "operation": "select",
        "argument": "worker (279472)"
      },
      {
        "operation": "relate",
        "argument": "wearing, subject, Personal protective Equipment (271881)"
      },
      {
        "operation": "exist",
        "argument": "?"
      }
    ],
    "semanticStr": "select: worker (279472) -> relate: wearing, subject, Personal protective Equipment (271881) -> exist: ?"
  }
}
```

Table 4

Results of construction entity detection.

Metric	Mean	Worker	Safety helmet	Scaffold	Safety harness	Guardrail	Bulldozer	Excavator
Precision	0.891	0.956	0.914	0.895	0.847	0.826	0.841	0.962
Recall	0.830	0.785	0.814	0.845	0.784	0.832	0.851	0.899

ViT. The evaluation dataset for ViT was produced by [47]. The results are promising in common construction object detection, which are robust enough for the semantic encoding process.

In the context of ITR, the goal is to find the most relevant images given keywords. Therefore, the performance of ITR can be assessed by determining how well it retrieves the correct images. In the context of ITR evaluation, we re-defined TP and FP. A TP is an instance where the relevant image is retrieved for a given keyword, while a FP is when the relevant image is not retrieved. We employed multiple image-wise recall metrics, namely Recall@5, Recall@10, Recall@50, and Recall@100, to comprehensively assess the system performance. These metrics calculate the percentage of top X confident relevant predictions that are accurate for each keyword. The evaluation involved 7 keywords linked to distinct safety considerations. As shown in Table 5, “Scaffolding” has the best recall rates across all 4 metrics, indicating that the model is highly effective in retrieving relevant results for this keyword. Conversely, the keyword “Smoking” demonstrates the lowest recall rates among the evaluated terms. This underperformance might be attributed to the image encoder. ViT may be struggled to effectively detect small objects like cigarettes or lighters within images. These objects might not be adequately represented in the encoded patches, posing challenges for the model in accurately identifying and retrieving pertinent results.

A total of 178 questions were designed for VQA evaluation. The question types include number-related questions, danger type questions, and recommendation questions. Number-related questions pertain to queries involving numerical values or quantifiable information, such as the number of workers and other construction objects on site. These questions are essential in ensuring that safety standards are met and maintained throughout the construction process. Danger type questions are those that address potential hazards or risks associated with the construction site, equipment, or tasks. These questions are crucial for identifying and mitigating threats to the safety and well-being of construction workers and others present at the site. Recommendation questions focus on seeking suggestions or advice regarding safety procedures, equipment, or best practices. These questions play a vital role in improving safety measures and reducing the likelihood of accidents or injuries on the construction site.

As shown in Table 6, the VQA module demonstrates a relatively strong performance in handling number-related questions, while its accuracy in addressing danger type and recommendation questions is slightly lower. This could be due to a few factors:

1. Lack of training data: The VQA module may not have been trained on enough question examples to accurately answer them. Increasing the amount of training data for this type of question could help improve the model performance.
2. Ambiguity in the questions: Danger type and recommendation questions may be more likely to have ambiguous phrasing or require more specific information than number-related questions. This could

Table 5

Recall rates for ITR.

Keywords	Height Work	Smoking	Scaffolding	Excavation	PPE	Site Lighting	Safety Exits	Mean
Recall@5	0.856	0.574	0.864	0.845	0.821	0.832	0.815	0.801
Recall@10	0.897	0.593	0.894	0.854	0.864	0.879	0.861	0.835
Recall@50	0.926	0.611	0.916	0.888	0.902	0.905	0.896	0.863
Recall@100	0.942	0.657	0.957	0.897	0.916	0.915	0.909	0.885

Table 6

Results for VQA.

	Total Questions	Unanswerable Questions	Answerable But Wrong	Correct Answers
Number-related questions	55	5	10	40
Danger type questions	74	11	7	56
Recommendation questions	49	8	5	36

make it more difficult for the VQA module to accurately answer them.

3. Complexities in the visual data: Answering danger type and recommendation questions requires more abstract or context-dependent concepts associated with the capabilities of text-decoder.

Table 7 summarises the VQA accuracy in identifying some unsafe situations on a construction site based on input images and corresponding text descriptions. The module demonstrated robust performance, particularly in detecting workers not wearing safety vests or high-visibility harnesses, achieving high accuracy rates of 89.8% and 91.5%, respectively. It also performed well in identifying workers working in areas with inadequate lighting, with an accuracy of 92.7%. However, the module showed a slightly lower accuracy rate of 84.6% in identifying workers not wearing safety helmets. The VQA module demonstrated an average accuracy rate of 89.7%. It is important to note that VQA accuracy may be influenced by factors such as occlusion conditions and camera angles.

Semantic propositional image caption evaluation (SPICE) [3] was used to measure the IC performance. Rather than focusing on the individual words or n-grams in the generated captions, SPICE concentrates on the semantic meaning conveyed by the captions. To compute the SPICE score, the candidate caption and the reference captions are first parsed into scene graphs. Scene graphs are structured representations that capture objects, attributes, and relationships between objects in the captions. These scene graphs are then compared, and the similarity between the candidate and reference scene graphs is calculated.

Table 7

Accuracy of VQA for identifying unsafe situations.

Item	Text description of unsafe situations	Identification accuracy
1	Worker not wearing a safety vest while working on a scaffold.	89.8%
2	Worker not wearing a high-vis harness while working near an excavator.	91.5%
3	Worker working in an area with inadequate lighting.	92.7%
4	Worker working at a height without a safety helmet.	84.6%

The similarity is measured using F1-score, which considers both precision (how many of the elements in the candidate scene graph are present in the reference scene graph) and recall (how many of the elements in the reference scene graph are present in the candidate scene graph). The F1-score ranges from 0 to 1, with 1 being a perfect match between the candidate and reference scene graphs. F1-score is formulated as:

$$F1 \text{ Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100\% \quad (4)$$

To implement the SPICE metric, we used its official evaluation code, which requires the Java Development Kit. The code takes the candidate and reference captions, converts them into scene graphs, and performs the comparison. This method is computationally intensive due to the complexity of the scene graph parsing and semantic evaluation involved. For the IC evaluation, we used a benchmark dataset produced

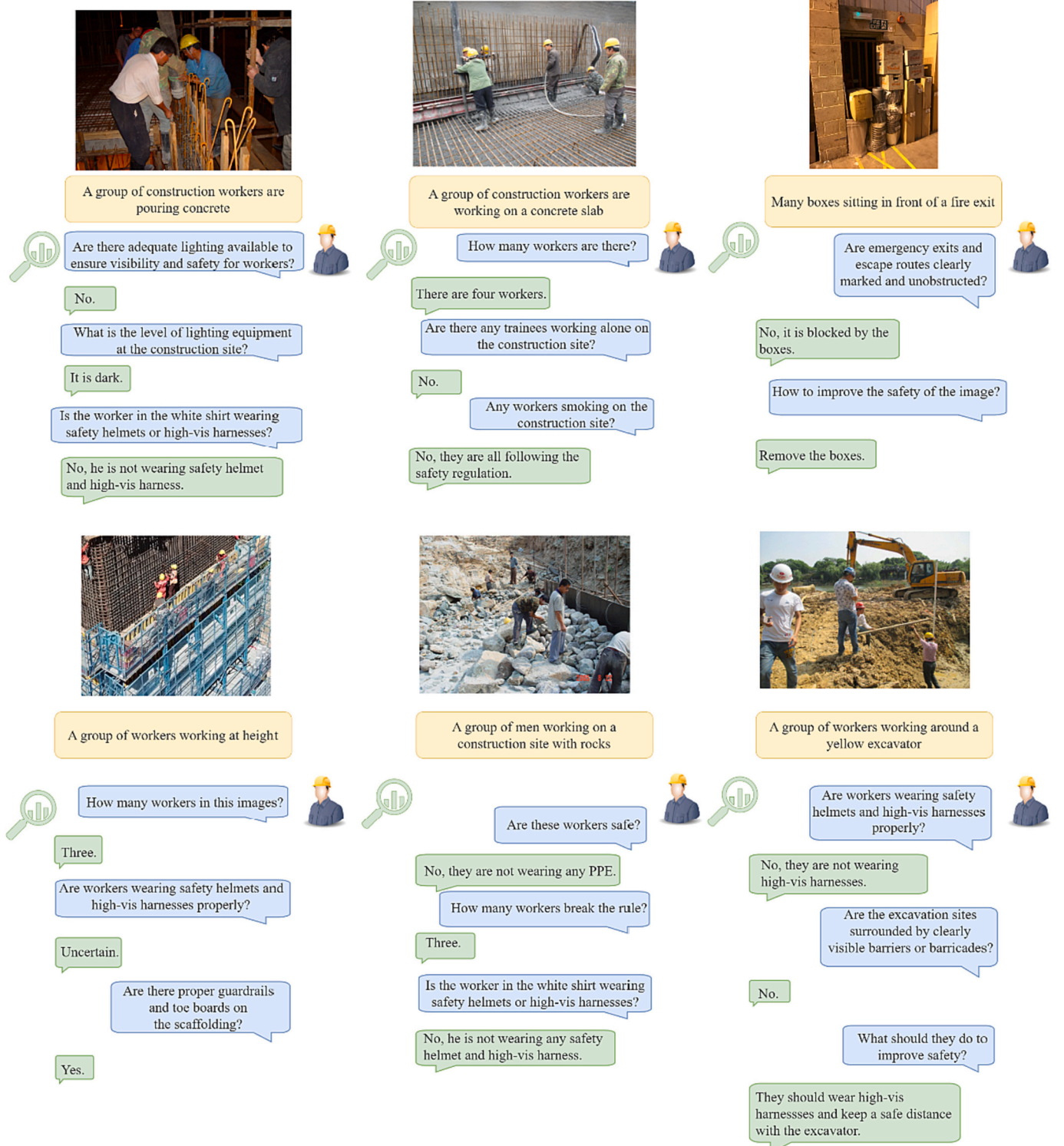


Fig. 6. Qualitative illustrations of the chat log and captions for various construction images.

by [48]. This dataset includes around 4000 images with 8000 captions, spanning a wide array of construction activities. We manually excluded 369 images based on the safety considerations specified for this study. Following the evaluation, we obtained a SPICE score of 0.449. It is essential to note that the SPICE score of 0.449 is a measure of semantic alignment between the generated captions and reference captions, rather than a straightforward accuracy metric. The SPICE score quantifies how well our model captures objects, attributes, and relationships in the images, making it a useful measure for assessing the model semantic understanding. Given the scale of 0 to 1, a score of 0.449 is significant for a proof-of-concept system, indicating its potential for effective semantic understanding in complex construction environments. This aligns our model with advanced systems in terms of its ability to extract semantic information.

Fig. 6 illustrates qualitative results from different construction scenarios, demonstrating the system capabilities in IC and VQA. For each depicted construction site image, corresponding image captions are provided to give an overview of the scene, and a visual dialogue between the system and the user is displayed. Specifically, text in yellow boxes signifies generated captions, blue boxes contain user queries, and green boxes show the system responses. The system has been designed to exercise caution, resorting to an ‘uncertain’ response when it cannot provide a confident answer. This is a safety feature that reduces the risk of

circulating inaccurate information. While it may not surpass human experts in every capacity, the system provides immediate, consistent feedback to support enhanced situational awareness among construction workers. By serving as an additional information layer, the system aids workers in making more informed decisions, thereby improving overall safety and efficiency on construction sites.

5. Preliminary test of the VCSQ system in construction scenarios

To better understand the system effectiveness and user perceptual capabilities, we conducted a preliminary test involved inviting ten construction practitioners who are already familiar with the three construction scenarios: progress monitoring, safety inspection and compliance checking. The participants were asked to use the AR-based VCSQ system to complete specific workflows related to these construction scenarios. After completing these tasks, participants were then required to fill out a survey. This survey was comprehensive, incorporating both the NASA-Task Load Index (NASA-TLX) and a section for open-ended comments. The NASA-TLX section collected data on their experiences using both AR and non-AR methods, with the non-AR data providing a benchmark of their typical performance in these construction activities. The inclusion of open-ended comments further enriched our understanding of the system practical applicability and highlighted potential

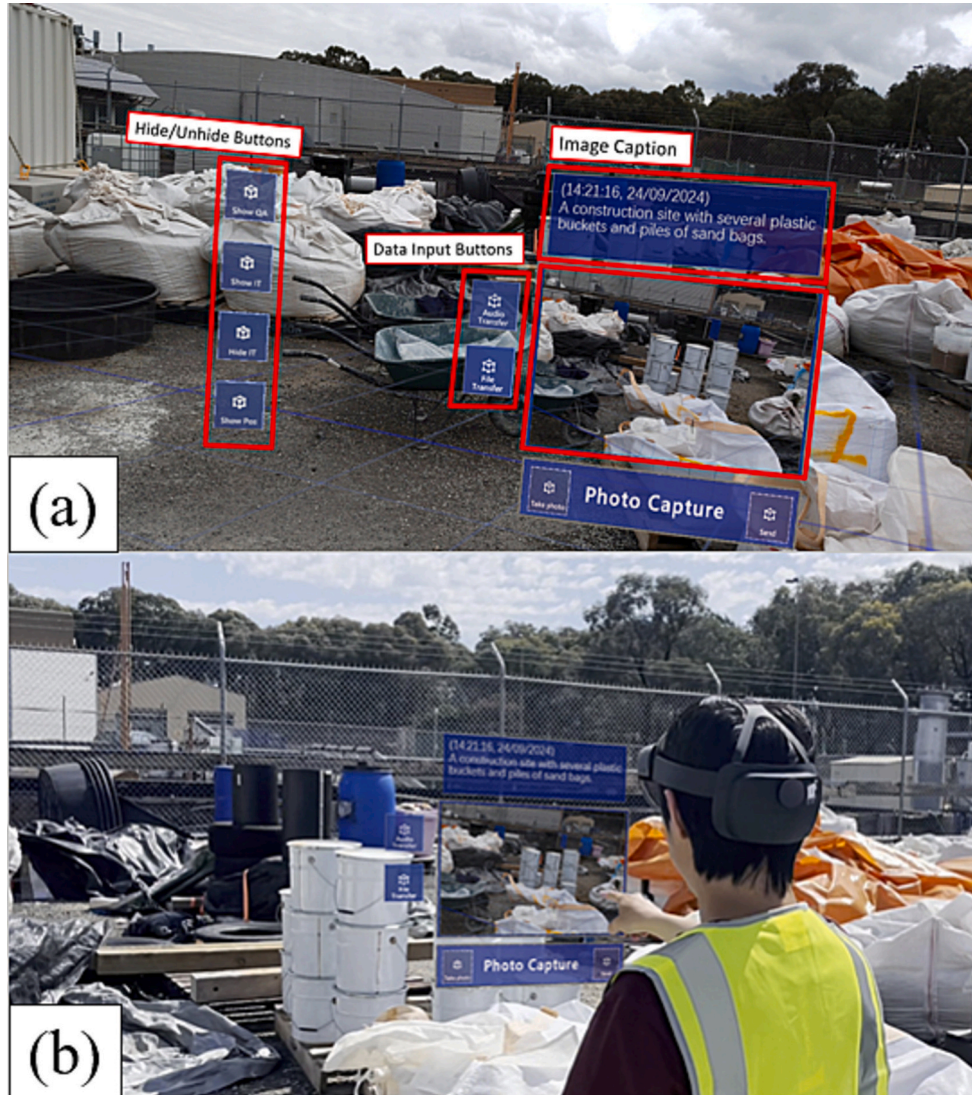


Fig. 7. UI of the IC module: (a) first person view; (b) third person view.

areas for improvement.

5.1. Scenario 1: Progress monitoring

In an experimental setup focused on progress monitoring in real-world construction contexts, participants engaged with the IC module via the AR interface in time-sensitive construction environments. Captured on-site images, as demonstrated in Fig. 7, triggered the IC module through a designated AR button. Initial results indicated that the module efficiently generated descriptive captions for these images. A notable experimental finding was the system ability for instant report creation, facilitated by an integrated data aggregation algorithm. The algorithm organised captured images, time stamps, and captions into a spreadsheet report, which was securely sent to managerial terminals through TCP protocols for immediate evaluation.

The one-touch report generation was subjected to two key performance indicators: administrative efficiency and situational awareness enhancement. Preliminary data suggested that the feature streamlined admin tasks while providing a real-time site overview. Further, participants employed the AR voice input to append action items to captured images, enabling immediate, data-driven construction management decisions.

As shown in Fig. 8, we used the NASA-TLX index to gauge workload and user experience. When compared to the non-AR method, which refers to the conventional workflow that site managers follow in construction settings, the AR method generally demonstrated lower mental and physical demand. Time pressure and stress levels were also notably less in the AR setting, and there was a slight improvement in the sense of task accomplishment. The reduced levels in the AR method imply that the one-touch reporting mechanism contributes to improved time management. Some participants find the one-touch report generation feature extremely convenient, as it removes the need for manual on-site report compilation. However, they also point out that the angle at which the AR glasses capture photos can substantially affect the accuracy of the IC module, leading to potential inaccuracies in the final reports.

5.2. Scenario 2: Safety inspection

In the context of safety inspection, the VQA module was tested in intricate and potentially hazardous construction settings, as evidenced in Fig. 9. A participant was assigned the task of inspecting an outdoor gas storage area. He engaged with the VQA module for guidance. Upon uploading an image of the storage area into the system, he received comprehensive descriptions that assisted in identifying crucial elements and related risks. For example, when the query “What is it?” was input, the system identified a secured area with multiple warning signs. A subsequent query from participants was, “How do I inspect this area?” The system responded with a structured inspection procedure that included assessments of the fence, state of containers, and other

equipment within the designated area. The final question concerned PPE suitable for the environment. System recommendations were consistent with identified hazards and included hard hats, safety glasses or goggles, and gloves. While the system may not supersede the acumen of seasoned professionals, initial findings suggest that it acts as a valuable supplementary tool. Participants utilising AR headsets equipped with the VQA module were better equipped to validate their risk assessments, alleviate uncertainties, and make well-informed, on-the-spot decisions. The experiment showed that the module enhances situational awareness and decision-making capabilities without intending to replace human expertise.

As shown in Fig. 10, the AR method generally outperformed the non-AR method in terms of mental and physical demand, as well as time pressure. In these aspects, participants found the AR setup less taxing and better for time management. Regarding task accomplishment and effort levels, both AR and non-AR methods showed similar levels of effectiveness. This supports the notion that the AR technology serves more as a supplement to human expertise rather than a replacement. Stress levels were also slightly lower in the AR method compared to the non-AR method, indicating that the AR interface potentially contributes to a less stressful working environment. Some participants believe that while the current VQA module may not yet serve as a full replacement for seasoned professionals, its interactive question-answer format offers a novel approach. This interactivity can help workers maintain situational awareness, making it a valuable asset in complex and potentially hazardous environments. Some Participants, however, have pointed out that the text generated in the VQA module can obstruct the user view and that the display of text within the virtual panel is not very clear. This requires users to exert additional effort to read the augmented text.

5.3. Scenario 3: Compliance checking

In an experimental setup highlighted in Fig. 11, the effectiveness of the ITR module was assessed for its capacity to retrieve site-specific information rapidly and accurately. The participants, mainly safety managers, were responsible for assessing compliance with “work at height” regulations across a large construction area. They initiated this process through a keyword search within the system. The system, engineered for precise keyword matching, quickly produced relevant images from a comprehensive surveillance database. This is a deviation from traditional construction monitoring methods, which often involve sifting through less relevant data. In this case, the keyword search enabled participants to focus directly on areas featuring guardrails, thus eliminating the need to browse irrelevant images. Additionally, the system facilitated the work of safety managers by enabling them to quickly perform targeted compliance checks for specific scenarios, such as ensuring that workers at height were wearing safety harnesses and helmets. This is crucial as compliance requirements can vary depending on the construction setting. The tailored approach greatly improved the

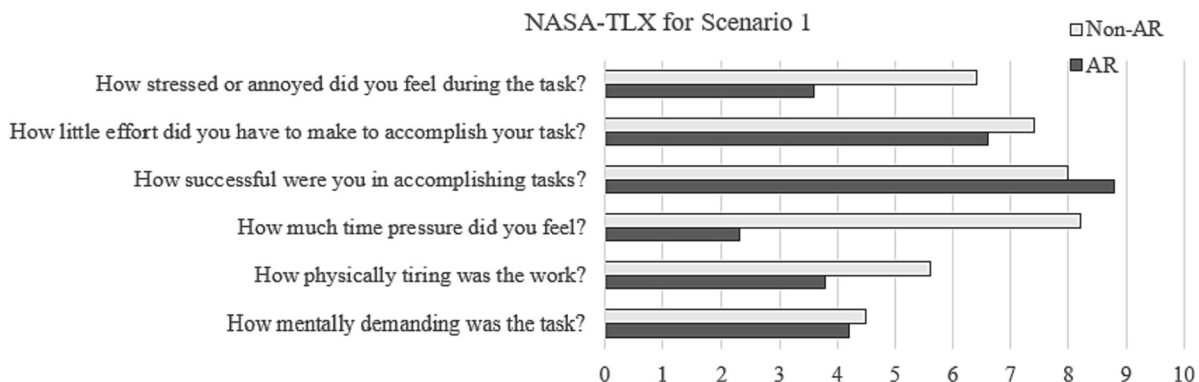


Fig. 8. NASA-TLX for construction progress monitoring.

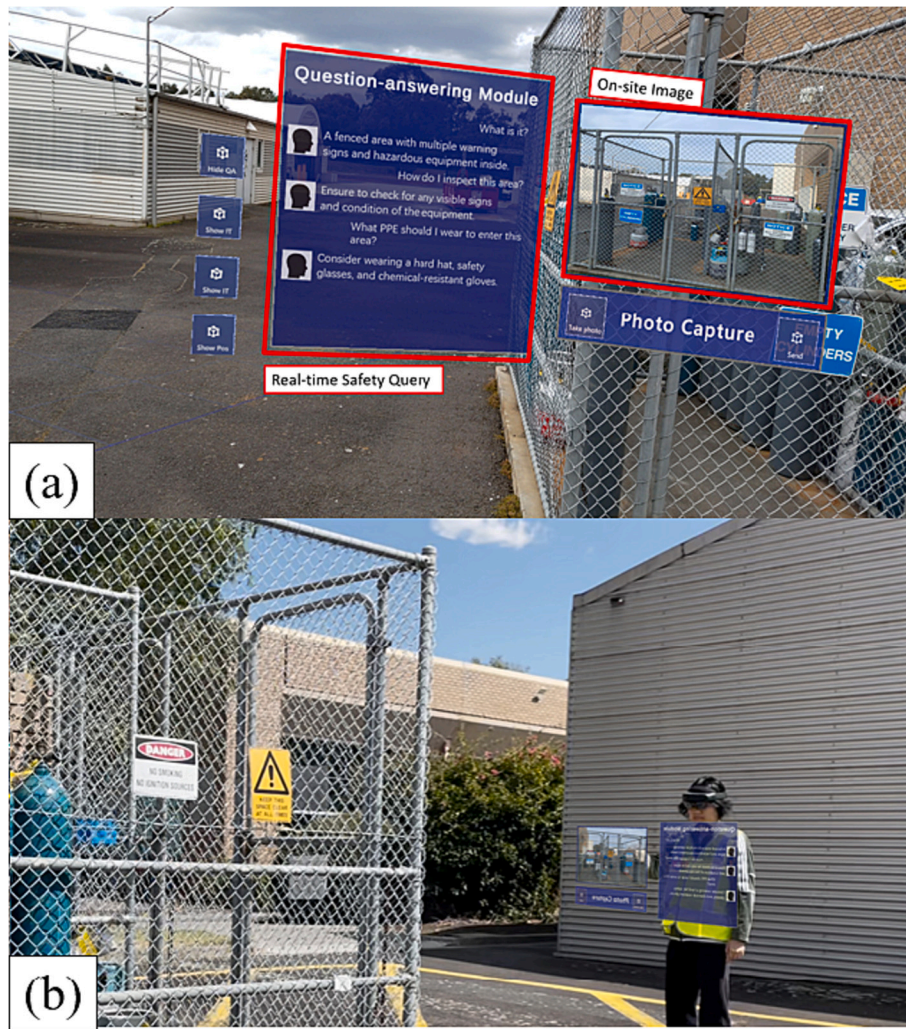


Fig. 9. UI of the VQA module: (a) first person view; (b) third person view.

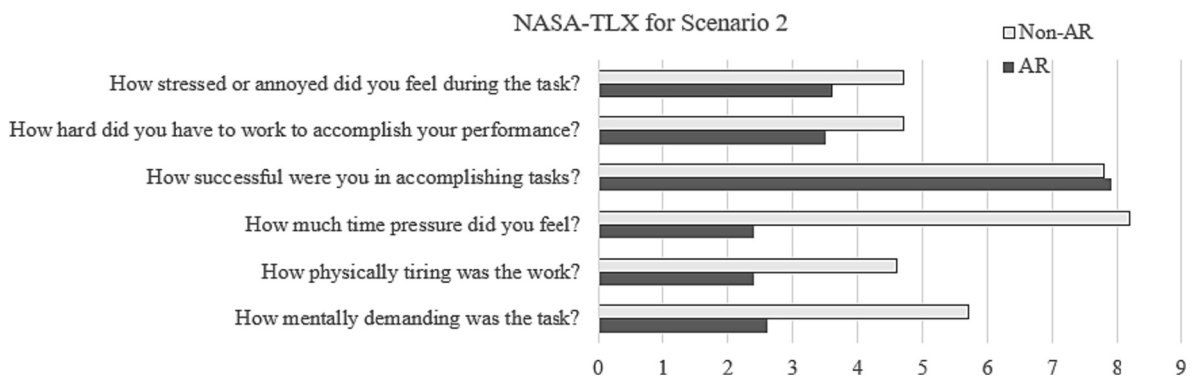


Fig. 10. NASA-TLX for real-time safety inspection.

efficiency of the compliance checks. This targeted approach generated specific data that could be assessed in real time, making it more effective than traditional surveillance or routine inspections.

As shown in Fig. 12, the NASA-TLX assessment revealed some notable insights. The AR environment was generally less mentally and physically demanding than the traditional non-AR methods. Participants in the AR scenario also reported significantly less time pressure, aligning well with the main findings that highlight the speed and accuracy of the ITR module. In terms of task success, both the AR and non-AR

environments proved effective, yet the AR scenario required significantly less effort to achieve similar results. Stress levels were comparable in both setups, indicating that the UI itself did not introduce new stress factors, despite being a novel interface.

Several participants have indicated that the use of the ITR module in an AR environment could substantially reduce the time and effort required to retrieve construction-related images. However, they also caution that the full deployment and implementation of this technology in actual construction sites may be challenged. One major hurdle is the



Fig. 11. UI of the ITR module: (a) first person view; (b) third person view.

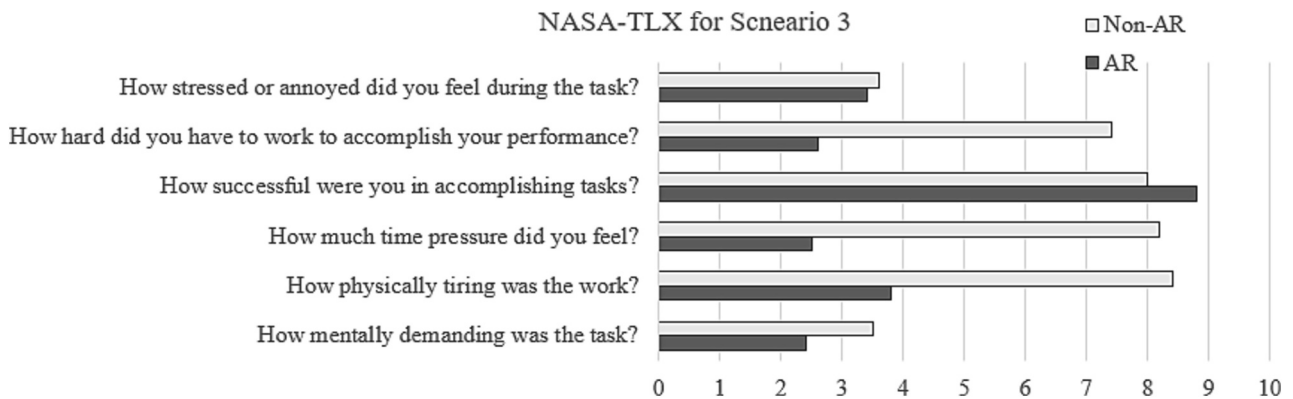


Fig. 12. NASA-TLX for safety compliance checking.

need for extensive camera setups to capture sufficient data, potentially including images of workers, equipment, and sensitive areas on the site. This raises questions about who has access to this data and how it is stored and used. Moreover, the constant surveillance capability of AR technology could be perceived as invasive, affecting worker morale and potentially creating a workplace culture of mistrust. This introduces potential ethical and moral dilemmas related to surveillance and privacy that would need to be carefully addressed.

6. Discussion

The VCSQ system is designed for construction practitioners, with the aim of enhancing construction safety and hazard mitigation. It serves a variety of functions, such as facilitating safety inspections and assisting in the generation of construction site reports. The system improves safety awareness by delivering real-time information on potential hazards and offers up-to-date data on the status of the construction site. The interactive nature of our system is also well-suited for in-field training.

An in-depth study that combines in-field training with transfer learning for real-time tasks could offer a comprehensive insight into how AR improves both learning and on-site safety in construction. Therefore, we recommend a comprehensive research agenda for the future that spans several crucial areas. These include initial safety induction before tasks commence, ongoing assessment mechanisms to gauge workforce performance, upskilling and reskilling initiatives to ensure employees are equipped with the latest safety knowledge, and post-task reflection sessions to evaluate what was learned and what needs improvement. For instance, a training scenario could be designed where trainee workers wear the AR glasses and navigate a simulated construction site. As they move through the environment, they could receive real-time prompts for identifying and annotating potential hazards. Each correct identification could trigger a contextual explanation, offering insights into best practices for mitigating the identified risks. This interactive training model would not only enhance hazard awareness but also offer immediate, experiential learning.

According to the system evaluation and participant feedback, the current system has certain limitations. First, our experiment scenarios are simulation-based. In real-world construction settings, factors like weather condition, equipment availability, and human error can change, evolve, or alter, potentially causing unexpected issues. Second, technological limitations are still diverse from the AR development perspective. AR glasses can obstruct the user field of view and their weight can contribute to fatigue. Moreover, the technology has the potential to introduce human error, as users may over-rely on the system and neglect their own safety assessments. The use of AR headsets can also cause physical discomfort and distractions, compromising focus on tasks at hand. In the short term, improvement of user-friendliness of system UIs could offer a more immediate solution. In the long term, it may be worthwhile for hardware vendors to focus on refining their products across the board. Third, the data set used for training our system is limited in scope, therefore, a richer data set is needed for fine-tuning the vision-language models. Fourth, integrating ontology with LLMs like GPT-4 [31] can potentially make the system more intelligent and capable of nuanced safety assessments. The embedding of safety regulations through ontology can address the feasibility issues posed by the absence of such standards in the existing dataset.

7. Conclusions

This study introduces the VCSQ system, utilising a unified vision-language model to offer functionalities like IC, VQA, and ITR. Paired with AR glasses, the system provides an interactive interface tailored for on-site workers and managers, thereby boosting safety awareness and work efficiency. Our technical contribution is the development of a unique prompt-tuning strategy that refines system responses in construction-specific safety contexts. This informs future integration of more advanced LLMs into construction safety. Our system, unlike generic LLMs, prioritises real-world applicability and customisation. The tangible benefits seen in our experimental results and practical applications suggest our system as a valuable tool for improving on-site situational awareness. This provides both researchers and practitioners a reference point in terms of interactive AI, potentially leading to incremental but important enhancements in current safety practices. Despite its promise, the system faces limitations, including technological constraints and the need for a richer dataset for model fine-tuning. For future refinements, the plan is to integrate an ontology-based framework that maps directly to established safety regulations in the construction industry. This will involve encoding these regulations into a structured data model that can be easily interpreted by the VCSQ system, thereby enhancing its role as a comprehensive compliance platform. Last but not least, research attention must extend beyond the technical realm to encompass challenges associated with the human aspect of technology adoption. This includes investigating how well the system is received by end-users and devising appropriate training modules to facilitate its

effective utilisation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Ministry of Housing and Urban Rural Development of China (MHURDC), Vol. 2023, <http://www.mohurd.gov.cn>.
- [2] Work-related traumatic injury fatalities Australia Vol. 2023, 2021. <https://www.sa.gov.au/doc/work-related-traumatic-injury-fatalities-australia-2021>.
- [3] P. Anderson, B. Fernando, M. Johnson, S. Gould, Spice: Semantic propositional image caption evaluation, in: Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part V 14, Springer, 2016, pp. 382–398, https://doi.org/10.1007/978-3-319-46454-1_24.
- [4] S. Bang, H. Kim, Context-based information generation for managing UAV-acquired data using image captioning, *Autom. Constr.* 112 (2020), 103116, <https://doi.org/10.1016/j.autcon.2020.103116>.
- [5] A.H. Behzadan, V.R. Kamat, Enabling discovery-based learning in construction using telepresent augmented reality, *Autom. Constr.* 33 (2013) 3–10, <https://doi.org/10.1016/j.autcon.2012.09.003>.
- [6] M.D.O. Carrasco, P.-H. Chen, Application of mixed reality for improving architectural design comprehension effectiveness, *Autom. Constr.* 126 (2021), 103677, <https://doi.org/10.1016/j.autcon.2021.103677>.
- [7] H. Chen, L. Hou, G. Zhang, Social distance monitoring of site workers for COVID-19 using context-guided data augmentation, deep learning, and homography transformation, in: IOP Conference Series: Earth and Environmental Science vol. 1101, IOP Publishing, 2022, p. 032035, <https://doi.org/10.1088/1755-1315/1101/3/032035>.
- [8] H. Chen, L. Hou, G.K. Zhang, S. Moon, Development of BIM, IoT and AR/VR technologies for fire safety and upskilling, *Autom. Constr.* 125 (2021), 103631, <https://doi.org/10.1016/j.autcon.2021.103631>.
- [9] H. Chen, L. Hou, G.K. Zhang, S. Wu, Using context-guided data augmentation, lightweight CNN, and proximity detection techniques to improve site safety monitoring under occlusion conditions, *Saf. Sci.* 158 (2023), 105958, <https://doi.org/10.1016/j.ssci.2022.105958>.
- [10] S. Chen, K. Demachi, Towards on-site hazards identification of improper use of personal protective equipment using deep learning-based geometric relationships and hierarchical scene graph, *Autom. Constr.* 125 (2021), 103619, <https://doi.org/10.1016/j.autcon.2021.103619>.
- [11] Y.-J. Chen, Y.-S. Lai, Y.-H. Lin, BIM-based augmented reality inspection and maintenance of fire safety equipment, *Autom. Constr.* 110 (2020), 103041, <https://doi.org/10.1016/j.autcon.2019.103041>.
- [12] H.-L. Chi, M.-K. Kim, K.-Z. Liu, J. Thedja, J. Seo, D.-E. Lee, Rebar inspection integrating augmented reality and laser scanning, *Autom. Constr.* 136 (2022), 104183, <https://doi.org/10.1016/j.autcon.2022.104183>.
- [13] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-Training of Deep Bidirectional Transformers for Language Understanding, *arXiv Preprint*, [arXiv:1810.04805](https://arxiv.org/abs/1810.04805), 2018, <https://doi.org/10.48550/arXiv.1810.04805>.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, *arXiv Preprint arXiv:2010.11929*, 2020, <https://doi.org/10.48550/arXiv.2010.11929>.
- [15] R. Eiris, M. Gheisari, B. Esmaeili, PARS: using augmented 360-degree panoramas of reality for construction safety training, *Int. J. Environ. Res. Public Health* 15 (11) (2018) 2452, <https://doi.org/10.3390/ijerph15112452>.
- [16] W. Fang, L. Ma, P.E. Love, H. Luo, L. Ding, A. Zhou, Knowledge graph for identifying hazards on construction sites: integrating computer vision with ontology, *Autom. Constr.* 119 (2020), 103310, <https://doi.org/10.1016/j.autcon.2020.103310>.
- [17] A. Harikrishnan, A.S. Abdallah, S.K. Ayer, M. El Asmar, P. Tang, Feasibility of augmented reality technology for communication in the construction industry, *Adv. Eng. Inform.* 50 (2021), 101363, <https://doi.org/10.1016/j.aei.2021.101363>.
- [18] L. Hou, H. Chen, G. Zhang, X. Wang, Deep learning-based applications for safety management in the AEC industry: a review, *Appl. Sci.* 11 (2) (2021) 821, <https://doi.org/10.3390/app11020821>.
- [19] L. Huang, W. Wang, J. Chen, X.-Y. Wei, Attention on attention for image captioning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 4634–4643, <https://doi.org/10.1109/ICCV.2019.00473>.
- [20] I. Jeelani, K. Han, A. Albert, Development of virtual reality and stereo-panoramic environments for construction safety training, *Eng. Constr. Archit. Manag.* 27 (8) (2020) 1853–1876, <https://doi.org/10.1108/ECAM-07-2019-0391>.

- [21] W. Jin, Y. Cheng, Y. Shen, W. Chen, X. Ren, A good Prompt is Worth Millions of Parameters? Low-Resource Prompt-Based Learning for Vision-Language Models, arXiv preprint arXiv:2110.08484, 2021, <https://doi.org/10.48550/arXiv.2110.08484>.
- [22] S. Joshi, M. Hamilton, R. Warren, D. Faucett, W. Tian, Y. Wang, J. Ma, Implementing virtual reality technology for safety training in the precast/prestressed concrete industry, *Appl. Ergon.* 90 (2021), 103286, <https://doi.org/10.1016/j.apergo.2020.103286>.
- [23] J. Kim, S. Chi, Graph neural network-based propagation effects modeling for detecting visual relationships among construction resources, *Autom. Constr.* 141 (2022), 104443, <https://doi.org/10.1016/j.autcon.2022.104443>.
- [24] J. Kim, J. Irizarry, Evaluating the use of augmented reality technology to improve construction management student's spatial skills, *Int. J. Constr. Educ. Res.* 17 (2) (2021) 99–116, <https://doi.org/10.1080/15578771.2020.1717680>.
- [25] K. Kim, H. Kim, H. Kim, Image-based construction hazard avoidance system using augmented reality in wearable device, *Autom. Constr.* 83 (2017) 390–403, <https://doi.org/10.1016/j.autcon.2017.06.014>.
- [26] J. Li, D. Li, C. Xiong, S. Hoi, Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, in: *International Conference on Machine Learning*, PMLR, 2022, pp. 12888–12900, <https://doi.org/10.48550/arXiv.2201.12086>.
- [27] Y. Li, H. Wei, Z. Han, N. Jiang, W. Wang, J. Huang, Computer vision-based hazard identification of construction site using visual relationship detection and ontology, *Buildings* 12 (6) (2022) 857, <https://doi.org/10.3390/buildings12060857>.
- [28] H. Liu, G. Wang, T. Huang, P. He, M. Skitmore, X. Luo, Manifesting construction activity scenes via image captioning, *Autom. Constr.* 119 (2020), 103334, <https://doi.org/10.1016/j.autcon.2020.103334>.
- [29] S. Moon, L. Hou, S. Han, Empirical study of an artificial neural network for a manufacturing production operation, *Oper. Manag. Res.* (2022) 1–13, <https://doi.org/10.1007/s12063-022-00309-0>.
- [30] O.R. Ogunseju, J. Olayiwola, A.A. Akanmu, C. Nnaji, Digital twin-driven framework for improving self-management of ergonomic risks, *Smart Sustain. Built Environ.* 10 (3) (2021) 403–419, <https://doi.org/10.1108/SASBE-03-2021-0035>.
- [31] OpenAI, GPT-4 Technical Report, ArXiv abs/2303.08774, 2023, <https://doi.org/10.48550/arXiv.2303.08774>.
- [32] Z. Pan, C. Su, Y. Deng, J. Cheng, Image2Triplets: a computer vision-based explicit relationship extraction framework for updating construction activity knowledge graphs, *Comput. Ind.* 137 (2022), 103610, <https://doi.org/10.1016/j.compind.2022.103610>.
- [33] F. Pilati, M. Faccio, M. Gamberi, A. Regattieri, Learning manual assembly through real-time motion capture for operator training with augmented reality, *Proc. Manuf.* 45 (2020) 189–195, <https://doi.org/10.1016/j.promfg.2020.04.093>.
- [34] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, Learning transferable visual models from natural language supervision, in: *International Conference on Machine Learning*, PMLR, 2021, pp. 8748–8763, <https://doi.org/10.48550/arXiv.2103.00020>.
- [35] F.M. Rezende-Filho, L.J.S. da Fonseca, V. Nunes-Souza, G.D.S. Guedes, L.A. Rabelo, A student-centered approach for developing active learning: the construction of physical models as a teaching tool in medical physiology, *BMC Med. Educ.* 14 (1) (2014) 1–9, <https://doi.org/10.1186/1472-6920-14-189>.
- [36] Y. Shen, L. Heacock, J. Elias, K.D. Hentel, B. Reig, G. Shih, L. Moy, ChatGPT and other Large Language Models are Double-Edged Swords, 2023, p. 230163, <https://doi.org/10.1148/radiol.230163>.
- [37] H. Son, C. Kim, Integrated worker detection and tracking for the safe operation of construction machinery, *Autom. Constr.* 126 (2021), 103670, <https://doi.org/10.1016/j.autcon.2021.103670>.
- [38] R. Song, X. Chen, Z. Liu, H. An, Z. Zhang, X. Wang, H. Xu, Label Mask for Multi-Label Text Classification, arXiv preprint arXiv:2106.10076, 2021, <https://doi.org/10.48550/arXiv.2106.10076>.
- [39] D. Tian, M. Li, Q. Ren, X. Zhang, S. Han, Y. Shen, Intelligent question answering method for construction safety hazard knowledge based on deep semantic mining, *Autom. Constr.* 145 (2023), 104670, <https://doi.org/10.1016/j.autcon.2022.104670>.
- [40] H. Wang, C. Xu, J. McAuley, Automatic Multi-Label Prompting: Simple and Interpretable Few-Shot Classification, arXiv preprint arXiv:2204.06305, 2022, <https://doi.org/10.48550/arXiv.2204.06305>.
- [41] T.-K. Wang, J. Huang, P.-C. Liao, Y. Piao, Does augmented reality effectively foster visual learning process in construction? An eye-tracking study in steel installation, *Adv. Civil Eng.* 2018 (2018), <https://doi.org/10.1155/2018/2472167>.
- [42] E.M. Wetzel, C.B. Farrow, Active learning in construction management education: faculty perceptions of engagement and learning, *Int. J. Constr. Manag.* (2021) 1–9, <https://doi.org/10.1080/15623599.2021.1974684>.
- [43] S. Wu, L. Hou, H. Chen, G.K. Zhang, Y. Zou, Q. Tushar, Cognitive ergonomics-based augmented reality application for construction performance, *Autom. Constr.* 149 (2023), 104802, <https://doi.org/10.1016/j.autcon.2023.104802>.
- [44] S. Wu, L. Hou, G.K. Zhang, H. Chen, Real-time mixed reality-based visual warning for construction workforce safety, *Autom. Constr.* 139 (2022), 104252, <https://doi.org/10.1016/j.autcon.2022.104252>.
- [45] B. Xiao, Y. Wang, S.-C. Kang, Deep learning image captioning in construction management: a feasibility study, *J. Constr. Eng. Manag.* 148 (7) (2022) 04022049, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002297](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002297).
- [46] Z. Xu, N. Zheng, Incorporating virtual reality technology in safety training solution for construction site of urban cities, *Sustainability* 13 (1) (2020) 243, <https://doi.org/10.3390/su13010243>.
- [47] A. Xuehui, Z. Li, L. Zuguang, W. Chengzhi, L. Pengfei, L. Zhiwei, Dataset and benchmark for detecting moving objects in construction sites, *Autom. Constr.* 122 (2021), 103482, <https://doi.org/10.1016/j.autcon.2020.103482>.
- [48] P. Zhai, J. Wang, L. Zhang, Extracting worker unsafe behaviors from construction images using image captioning with deep learning-based attention mechanism, *J. Constr. Eng. Manag.* 149 (2) (2023) 04022164, <https://doi.org/10.1061/JCEMD4.COENG-12096>.
- [49] B. Zhong, H. Li, H. Luo, J. Zhou, W. Fang, X. Xing, Ontology-based semantic modeling of knowledge in construction: classification and identification of hazards implied in images, *J. Constr. Eng. Manag.* 146 (4) (2020) 04020013, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001767](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001767).
- [50] B. Zhong, H. Wu, H. Li, S. Sepasgozar, H. Luo, L. He, A scientometric analysis and critical review of construction related ontology research, *Autom. Constr.* 101 (2019) 17–31, <https://doi.org/10.1016/j.autcon.2018.12.013>.
- [51] B. Zhu, Y. Niu, S. Lee, M. Hur, H. Zhang, Debaised Fine-Tuning for Vision-language Models by Prompt Regularization, arXiv preprint arXiv:2301.12429, 2023, <https://doi.org/10.48550/arXiv.2301.12429>.